Design Project #1
Counter + Calculator

Microprocessor, Fall Semester, 2026

Soongsil University

Course Grading

- Lecture (2 credits)
  - Midterm Exam (40%)
  - Final Exam (40%)
  - Flipped Learning (5%)
  - Homework (10%)
  - Attendance (5%)
- Design (1 credit) – two projects (50%, 50%)
  - Report (40%)
  - Design output (60%)

Schedule

- Demonstration on Nov. 26.
- Final report due date: Dec. 21

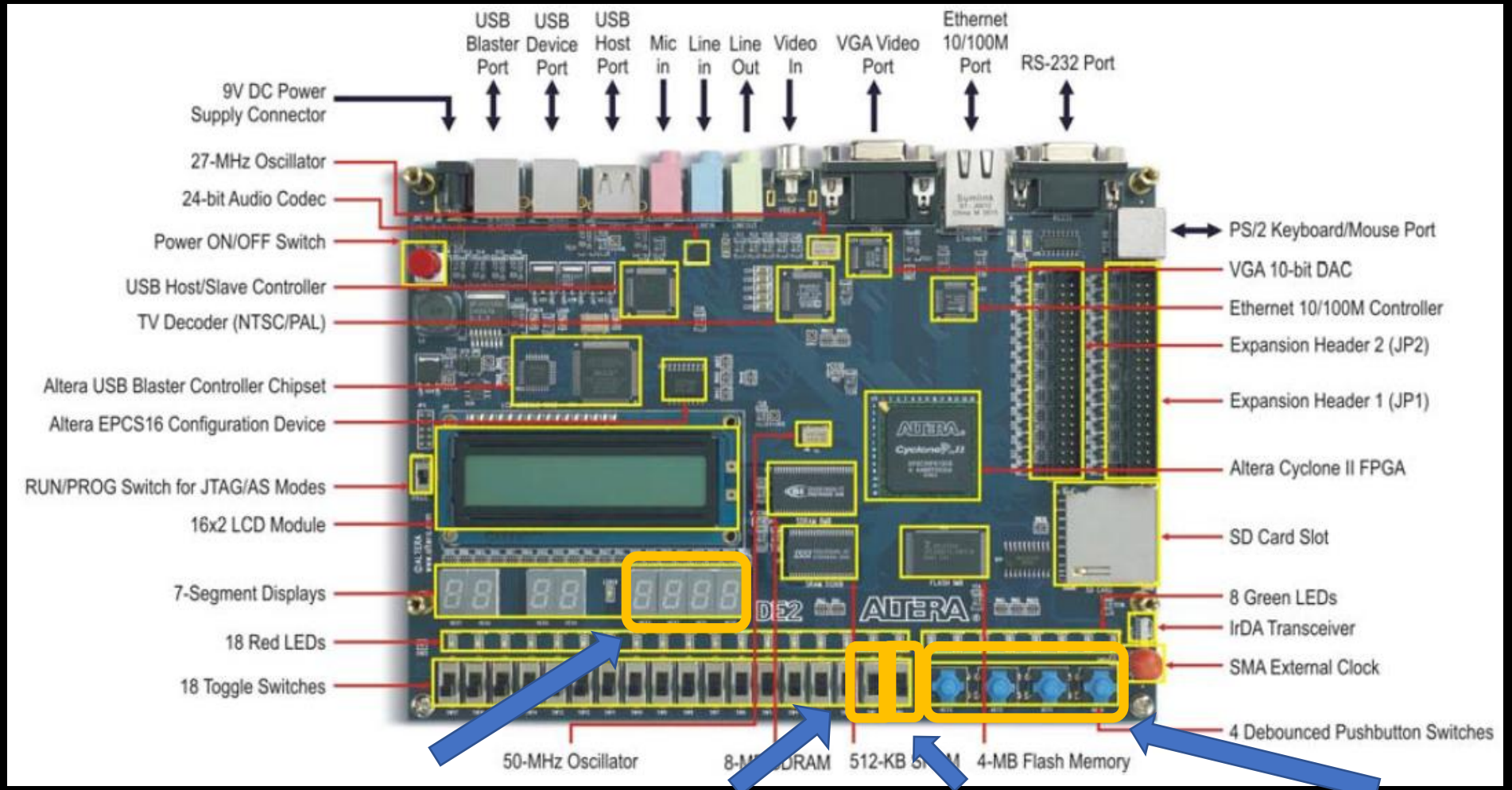
Counter



- Design a counter using DE2 board.
- Display the result to the 7-segment displays in decimal number form.
- Toggle switches define modes

T. Switch	Button 1	Button 2	Button 3	Button 4
0	Count one	Count two	Count down	Reset
1	Count five	Count ten	Count five down	Reset

DE2 Board Layout: Counter



Results

Counter mode

Counter/Calculator

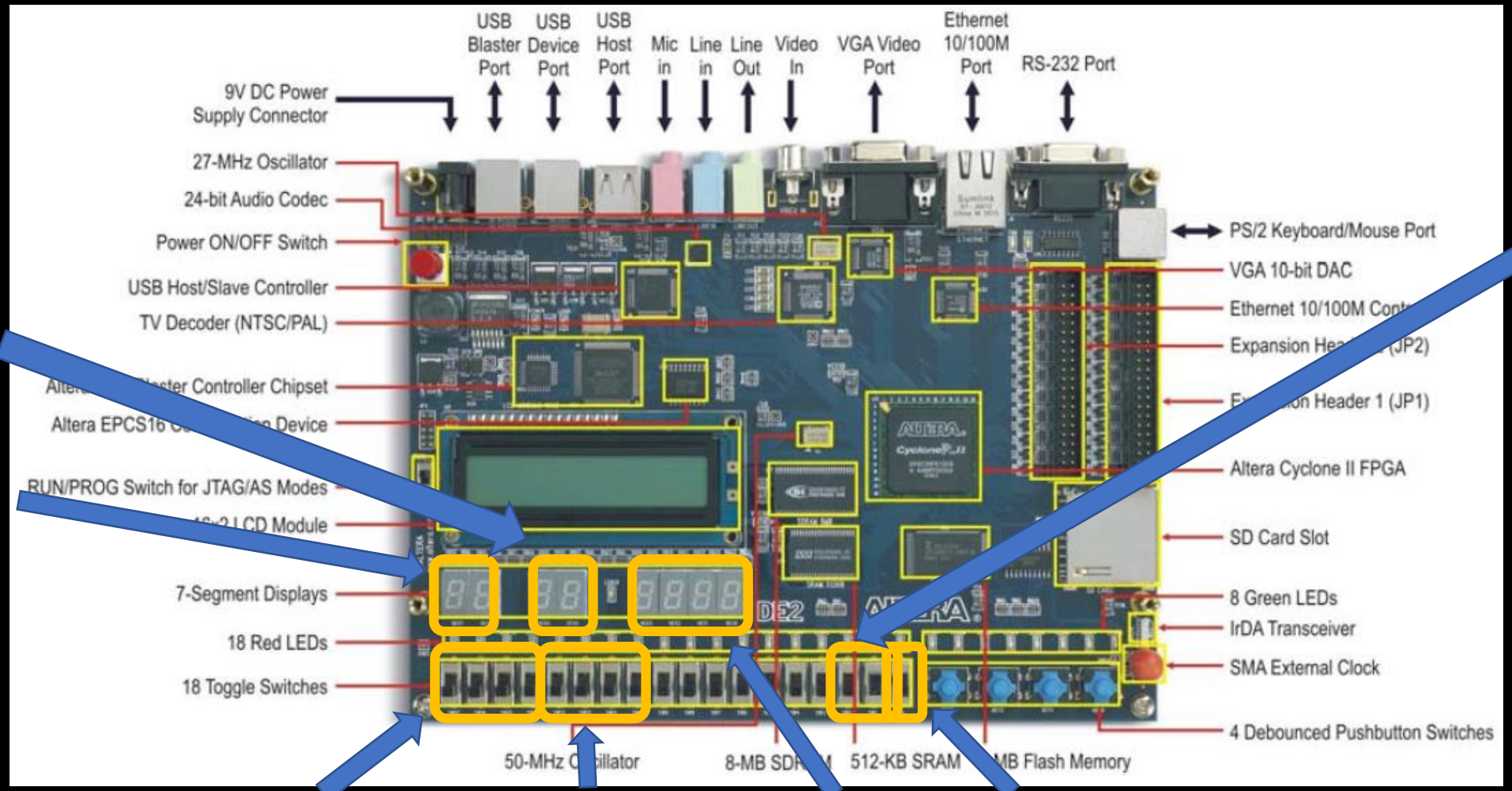
Action buttons

One Digit Calculator

- Receive a one-digit number from four toggle switches.
- Define the calculation type from toggle switches
  - 00: addition, 01: subtraction, 10: multiplication, and 11: division.
- Display the result on the 7-segment displays in decimal number form.

T. Switch	Calculation mode
00	Addition
01	Subtraction
10	Multiplication
11	Division

DE2 Board Layout: One Digit Calculator



Display the other number

Display one number

Calculator mode select

One number

The other number

Display result

Counter/Calculator

Evaluation Criteria

- Normal Counter Operation
  - Count up/down, display result, etc.
- Normal Arithmetic Operation
  - Addition, subtraction, multiplication, and division.
  - Exception, i.e., divided by 0.
- Readable and concise code. Minimize code lines as much as possible.

- Assessment of Understanding
  - Source code presentation.

Bonus Credit

- Counter
  - Additionally, display the result using LEDs. (+5)
  - Jackpot! When the counter result is 777, show something. (+5)

- Calculator
  - Remainder in division (+5)
  - Display in Hexadecimal number (+5)